

The story so far

None of us has a finance background, and the file is 230,111 rows. This page walks through everything in order: what's actually in each column, where this specific file will quietly mislead us, what a "hypothesis" even means for this assignment – and then the three real hypotheses we settled on and what the data actually showed when we checked them. Nothing here is fabricated. Every number below was pulled straight out of `stock_data.csv`, the same file all three of us are working from.

230,111

ROWS

49

COMPANIES

2006 → 2026

DATE SPAN

3

HYPOTHESES COMMITTED

01

Start here

This file has one row per company per trading day: a date, a ticker symbol, four prices, an adjusted price, and how many shares changed hands. That's it. No earnings, no news, no company names even – just **005930.KS** and a number.

That's simpler than it sounds, and also stricter than it sounds: it means every argument we make this semester has to be buildable out of price and volume alone. Keep that in the back of your mind as you read – we'll come back to exactly what that does and doesn't let us claim in section 11.

Reading one row

Take a single row: 2020-03-16, TSLA, 31.30, 32.99, 29.48, 29.67, 29.67, 307342500. Here's what each field means.

Ticker is the company's trading symbol. A suffix after a dot tells you which exchange it trades on – and that turns out to matter a lot (section 04):

.KS → Korea (KRW)

.HK → Hong Kong (HKD)

.SW → Switzerland (CHF)

.PA → Paris (EUR)

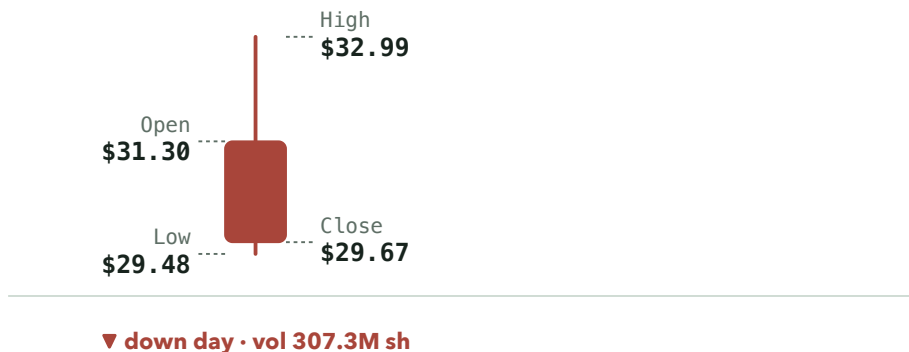
.SR → Saudi Arabia (SAR)

no suffix → US listing (USD)

A trading day isn't one price, it's thousands of trades happening continuously for hours. **Open, High, Low, Close** compress that whole day into four numbers: the first trade, the highest trade, the lowest trade, and the last trade. **Volume** is simply how many shares changed hands that day.

Anatomy of one real trading day

TSLA · March 16, 2020 · the week the market fell fastest during the COVID crash



Tesla opened the day at \$31.30, briefly traded as high as \$32.99 and as low as \$29.48, then closed at \$29.67 – down for the day. 307.3 million shares traded, far more than a calm day. One candle, four numbers, one volume bar.

Close vs. Adjusted Close

Two things happen to a stock's price that have nothing to do with the company getting more or less valuable: it can **split** (each share becomes several, each worth proportionally less) or it can pay a **dividend** (cash leaves the company and goes to shareholders, so the share is worth slightly less right after). **Adj Close** exists to undo both effects so you can measure real growth over time.

Here's the part most tutorials get wrong, and that we confirmed by checking real rows in this exact file:

What the public record says

AAPL, before its 4-for-1 split

Apple split its stock 4-for-1 on August 31, 2020. On **August 28, 2020**, the day before, its publicly reported closing price was roughly **\$499**.

A truly "raw, as-traded" feed would show ~\$499 for that date, forever – that's what actually printed on the tape that day.

What this file shows for the same day

Row: 2020-08-28, AAPL

Close = 124.81 Adj Close = 121.17

Both columns already reflect the split that hadn't happened yet on that date. This file rewrites the entire **Close** history every time a split occurs – not just **Adj Close**.

WHAT THIS MEANS FOR US

Don't assume **Close** is "the raw traded price" the way it might be elsewhere – here, it already isn't, for splits. The gap that's *left over* between **Close** and **Adj Close** (that 3% above) is dividends only. Rule of thumb: use **Adj Close** whenever you're computing a return or a growth rate; use **Close** only when you need the literal price someone would have paid on a given day.

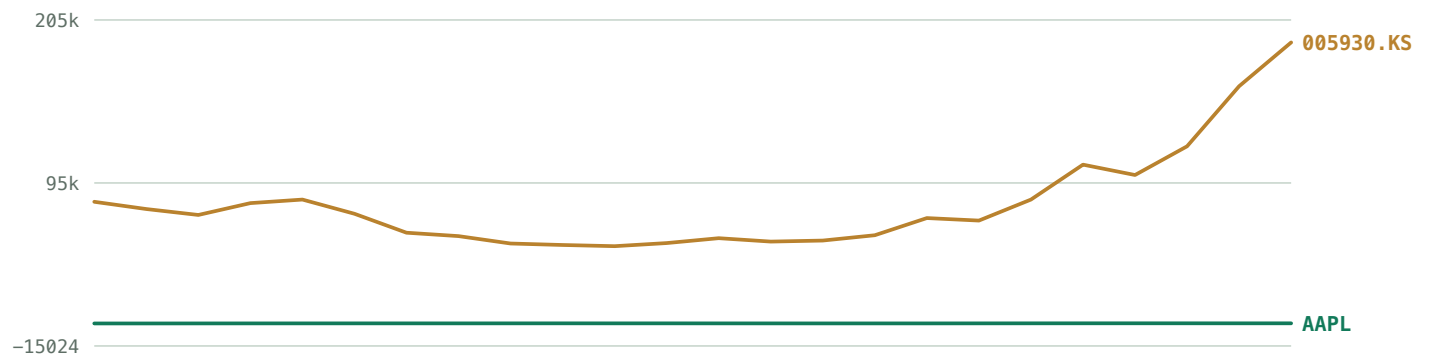
The currency trap

This file holds 49 companies from 6 different currencies, all sitting in the same **Close** column. Plot two of them together without thinking, and you get this:

The naive chart

AAPL vs. Samsung, raw Close, last 24 months

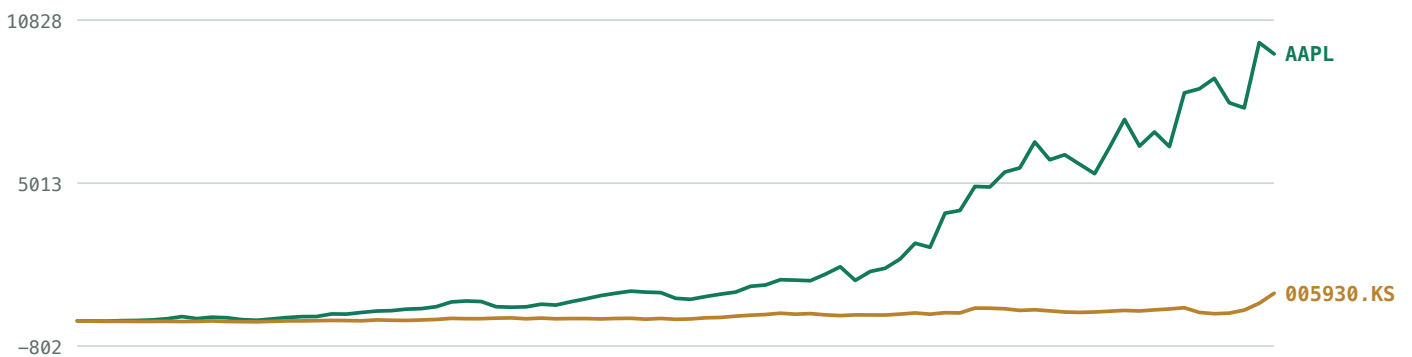
● AAPL (USD) ● 005930.KS (KRW)



The honest chart

Same two companies, indexed to 100 at 2006-01, full history

● AAPL ● 005930.KS



WHY THE FIRST ONE LIES

AAPL's last close in the file is **\$264.58**. Samsung's is **₩190,100**. Same day, same column, incomparable units – on a shared axis, AAPL is a flat line pinned near zero.

Whenever we compare tickers across the currency groups in section 02, we index each to 100 at a shared start date (or use percentage returns) instead of plotting raw price.

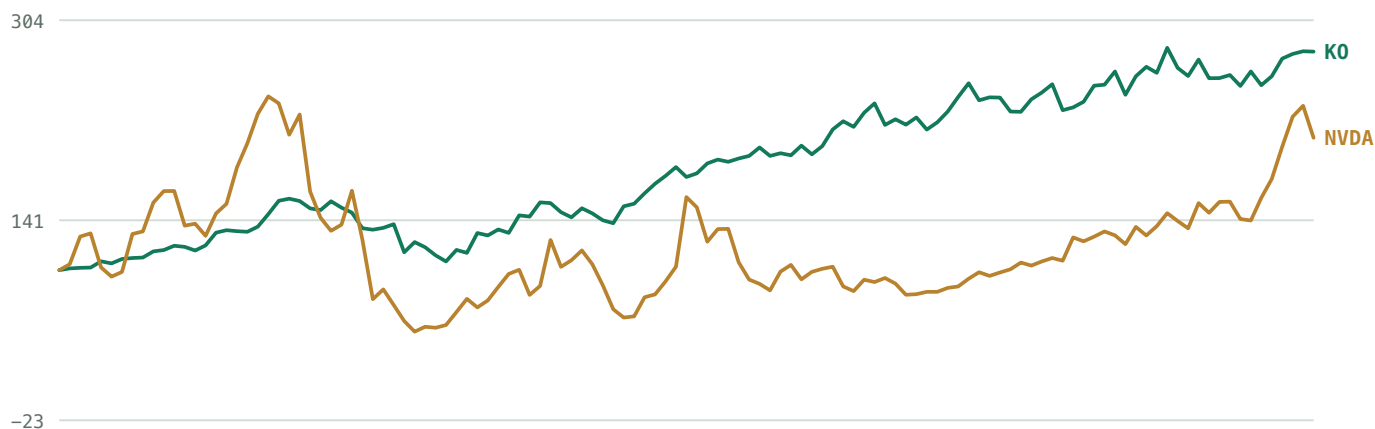
Returns & volatility

A **return** is just a percentage change: if a stock goes from \$100 to \$105, that's a 5% return. **Volatility** measures how much those returns bounce around – not how much money is lost, a common mix-up. A stock can be volatile and still end up way ahead; it just gets there on a much rougher path.

A steady mover vs. a jumpy one

KO and NVDA, indexed to 100 at 2006-01, ten-year window

● KO – Coca-Cola ● NVDA – Nvidia



Same starting point, same ten years. KO's path is a gentle, fairly predictable climb. NVDA's covers the same ground but whipsaws the whole way – measured as the standard deviation of monthly returns across the full 20-year file, NVDA's typical monthly swing (13.6%) is about three times the size of KO's (4.6%). That's what "volatile" actually means, numerically.

A WARNING ABOUT "LONG TERM"

Stretch this same comparison to the *full* 20-year file instead of just these ten years, and NVDA finishes about **550x** its starting value against KO's roughly **7x** – a gap so large the two lines can't even share a readable axis anymore. We're not cherry-picking to make a point; it's the opposite point. Change the end date and the entire story flips. That's exactly why a real hypothesis has to name a specific time window instead of saying "long term" – more on this in section 09.

Do markets move together?

Correlation is a single number from -1 to 1 describing whether two things tend to move up and down together. 1 means they move in perfect lockstep, 0 means no relationship, negative means they tend to move opposite each other. Below is the real correlation of monthly returns between five companies from five different exchanges, computed across the entire file.

Monthly-return correlation, 2006-2026

Five companies, five exchanges, full history

	AAPL	0700.HK	005930.KS	MC.PA	NOVN.SW
AAPL	1.00	0.31	0.29	0.46	0.17
0700.HK	0.31	1.00	0.10	0.30	0.01
005930.KS	0.29	0.10	1.00	0.25	0.14
MC.PA	0.46	0.30	0.25	1.00	0.19
NOVN.SW	0.17	0.01	0.14	0.19	1.00

Darker means more correlated. Even the strongest pair here – Apple and LVMH (Paris) at 0.46 – is far short of 1 . Averaged over 20 years, these markets are only loosely connected to each other.

THIS BECOMES H2, LATER ON THIS PAGE

"Averaged over 20 years" is doing a lot of work in that sentence. Does that looseness hold up during a crisis, or does it tighten? That question is exactly what turned into our second hypothesis – jump to section 11 if you want the answer now, or keep reading in order.

Your 49 companies

Grouped by the currency they trade in, and where they sit on the timeline.

CURRENCY	EXCHANGE	COMPANIES IN THIS FILE
USD	United States (incl. ADRs)	AAPL, ABBV, AMD, AMZN, ASML, AVGO, AZN, BABA, BAC, BRK-B, CAT, COST, CSCO, CVX, GE, GOOGL, HD, HSBC, JNJ, JPM, KO, LLY, LRCX, MA, META, MRK, MSFT, MU, NFLX, NVDA, ORCL, PG, PLTR, TM, TSLA, TSM, V, WMT, XOM
KRW	Korea (.KS)	000660.KS (SK Hynix), 005930.KS (Samsung)
HKD	Hong Kong (.HK)	0700.HK (Tencent), 0939.HK, 1288.HK, 1398.HK
CHF	Switzerland (.SW)	NOVN.SW (Novartis), ROG.SW (Roche)
EUR	Paris (.PA)	MC.PA (LVMH)
SAR	Saudi Arabia (.SR)	2222.SR (Saudi Aramco)



Five ways this data bites back

A quick-reference list – check this before trusting any chart that spans regions, eras, or tickers with different start dates.

01

Mixed currencies

Six currencies share one `Close` column. Index to 100 or use returns – never raw price across tickers. (Section 04)

02

Uneven start dates

Only 33 of 49 tickers span the full file. PLTR starts 2020-09-30, BABA 2014-09-19, META 2012-05-18, TSLA 2010-06-29. Rebase comparisons to a date every plotted ticker actually has.

03

Trading calendars don't line up

Full-history row counts differ by exchange – US 5,065 · Switzerland 5,070 · Paris 5,152 · Korea 4,972 · Hong Kong 4,964. Joining on date across regions leaves gaps.

04

`Close` already includes splits

The retroactive rewrite from section 03. What's left between `Close` and `Adj Close` is dividends only.

05

No fundamentals in this file

Only price and volume – no earnings, no revenue, no news. We can talk about co-movement and timing, never about *why*. (Section 11)

What is a hypothesis

In everyday speech, "hypothesis" just means a guess. For this assignment it means something narrower: **a specific, falsifiable claim about why the data looks the way it does** – something the data could actually embarrass you on, not a fact everyone already agrees with.

WEAK – A FACT, NOT A HYPOTHESIS

"Tech stocks went up a lot."

True, obvious, nobody disagrees. One groupby, one chart, nothing left to say.

STRONG – AN ACTUAL HYPOTHESIS

"The post-2023 rally is a narrow AI-infrastructure story, not a broad tech one — chipmakers detached from the rest of large-cap tech."

Names a mechanism, a time window, a group split. It implies a comparison – and it could turn out to be wrong. This is our real H1, not a made-up example – section 10 has the actual chart.

The four-part test

01

Falsifiable

You can say, in advance, what the chart would look like if you're wrong.

02

Non-obvious

If the reader already believes it, there's nothing to show them.

03

Specific

Named tickers, a named group, a named window – not "tech" or "long term."

04

Answerable

Provable with price and volume alone – see section 14 for what that rules out.

H1 • the AI rally is narrow

The claim: since 2023, "tech went up" is really "a handful of chip and AI companies went up a huge amount, and the rest of tech went up a normal amount." If you thought owning a broad basket of tech names gave you spread-out exposure to the AI story, the two groups below show that isn't really true – almost all of the extreme gain sits in one narrow group.

We split large-cap tech into two groups and averaged each one into a single line, indexed to 100 on 2023-01-03 (that just means: pretend every stock in the group was worth 100 on that day, so we can compare shape and speed of growth, not raw price):

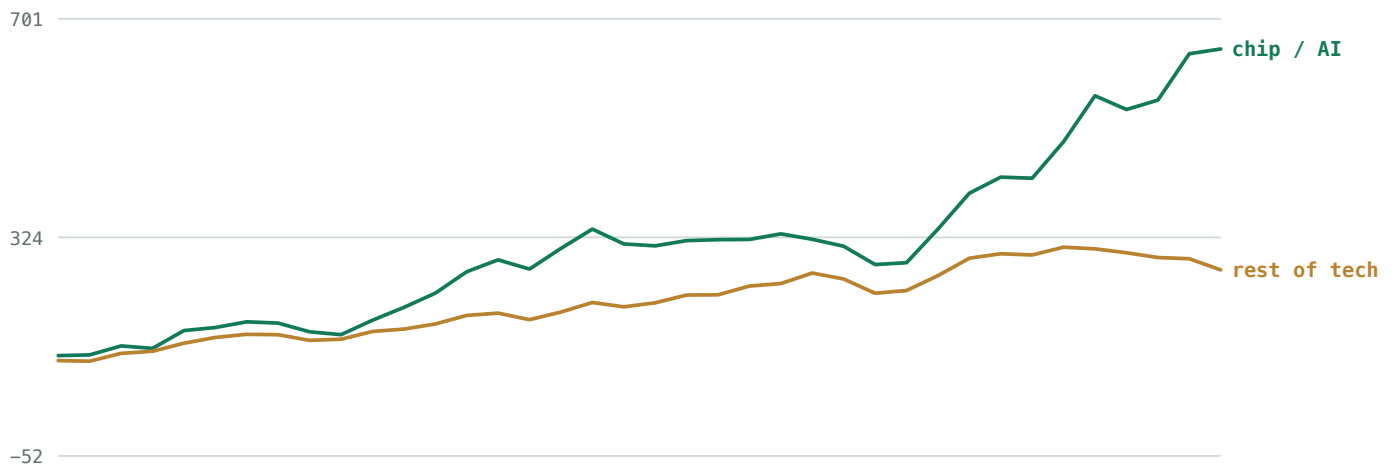
chip / AI group: NVDA, AVGO, TSM, ASML, AMD, MU, LRCX

rest of tech: AAPL, MSFT, GOOGL, AMZN, META, NFLX, ORCL, CSCO

Figure A – chip/AI group vs. rest of tech

Indexed to 100 on 2023-01-03, monthly, through 2026-02

● chip / AI group average ● rest of tech average



Both groups start at exactly 100 on the same day. The chip/AI group ends near 649 (a 549% gain on average); the rest-of-tech group ends near 268 (a 168% gain). The gap doesn't appear as one early jump – it keeps widening the whole way, month after month.

This isn't the first time we've seen this shape – Assignment 2's Figure 8 already showed NVDA and AVGO pulling away from ASML, TSM, and Apple since 2009, looking at individual stocks. This chart checks the same idea a different way: two whole groups, averaged, since 2023 specifically. Two different methods, same answer, so H1 holds up strongly.

H2 · diversification fails, but only in panics

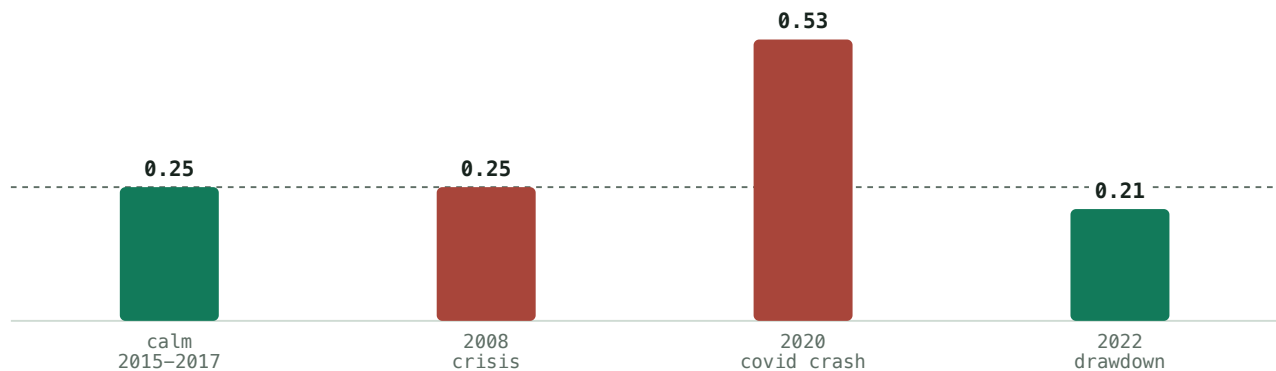
The claim: owning stocks from different regions of the world – US, Korea, Hong Kong, Switzerland, Paris – should protect you, because a bad day in one market shouldn't mean a bad day everywhere. That protection turns out to be real most of the time, and to disappear specifically during panic-style crashes.

We picked one stock per region – JPM (US), 005930.KS (Korea), 0700.HK (Hong Kong), NOVN.SW (Switzerland), MC.PA (Paris) – the same one-stock-per-region approach Assignment 2 used for LVMH representing Paris. For every pair of those five, we measured **correlation**: a score from -1 to 1 for how much two stocks move up and down on the same days. Close to 0 means they're moving pretty independently; close to 1 means they're moving almost in lockstep. We averaged that score across all ten pairs, in four different stretches of time.

■ calm / non-panic period ■ panic-style crash

Average cross-region correlation

All ten region pairs, averaged, four time windows



Calm years (2015–2017) sit around 0.25 – genuinely loose. The 2020 COVID crash jumps to 0.53 , more than double – when the whole world panics at once, region stops mattering and everything sells off together. 2022 actually stays lower than calm, at 0.21 , so this isn't "crashes always raise correlation" – it's specifically panic-style crashes.

THE HONEST CAVEAT, NOT SWEEPED UNDER THE RUG

2008 comes out at 0.25 – barely different from calm, even though 2008 was arguably the biggest global panic in the dataset. We don't think that means diversification

quietly worked in 2008. Our US stock for that window is JPM, and JPMorgan acquired the collapsing Bear Stearns in March 2008 – JPM's own trading volume ran roughly 5× normal that year for reasons that have nothing to do with global contagion. With only one stock standing in for an entire region, that single company's unusual year likely drowns out the real regional signal. We're flagging this as noisy data, not proof the pattern doesn't hold.

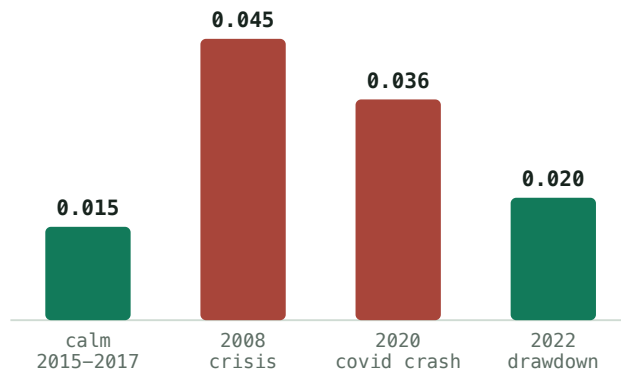
H3 · volume & volatility agree

The claim: if 2008 and 2020 were real panics and 2022 wasn't, that should show up in two other numbers entirely – how much prices swing around day to day (**volatility**) and how many shares are changing hands (**volume**). These don't depend on picking one stock per region the way H2 does, so they're a good independent check on the same story.

■ calm / non-panic period ■ panic-style crash

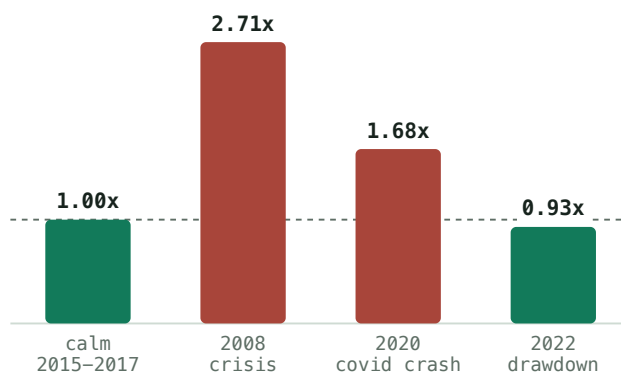
Avg. volatility

std. dev. of daily return



Avg. trading volume

vs. calm baseline



Volatility: 0.015 in calm years, then 0.046 in 2008 and 0.036 in 2020 – both real panics roughly double-to-triple the calm baseline – and only 0.020 in 2022, barely above

calm. Volume tells the same story: 1.0× normal in calm years, 2.7× in 2008, 1.7× in 2020, and actually slightly below normal (0.9×) in 2022.

WHY THIS MATTERS FOR H2

2008's volatility and volume numbers are both clearly elevated, exactly where H2's correlation number for 2008 looked suspiciously calm. That's useful: it lets us say "2008 was a real panic" with confidence from these two measurements, even while treating H2's own 2008 correlation figure as unreliable for the reason explained in section 11. Different measurements, same underlying story, and H3 is the one that isn't compromised by the one-stock-per-region limitation.

The story, all together

One hypothesis in isolation is still just a finding – a single chart, then you're done. A pile of unrelated hypotheses is a report. What makes a *story* is one central claim, followed by a chain of sub-claims, where each one answers the objection the previous chart raises. Here's our actual chain, built from H1, H2, and H3 above:

CENTRAL CLAIM

Diversification in this portfolio – across tech sub-sectors, or across world regions – is thinner than it looks.

OBJECTION

Surely a broad basket of tech names already spreads out the AI exposure?

H1

No – chip/AI names returned 549% since 2023 vs. 168% for the rest of large-cap tech. Almost all the extreme gain sits in one narrow group.

OBJECTION

Fine, but owning different **regions** is a completely different kind of spread – that should still protect you.

H2

It does, most of the time (correlation ~0.25, ~0.21). But in the 2020 panic it jumped to 0.53 – exactly when you'd most want that protection, it weakened.

OBJECTION

2008 was arguably a bigger panic than 2020, and its correlation number barely moved. Doesn't that undercut H2?

H3

Volatility and volume for 2008 are both clearly elevated (0.046 volatility, 2.7× volume) – it was a real panic, the correlation figure is just noisy because of JPM's own unusual year. Independent measurements back up the story where H2's own number can't.

THE PART NOBODY TELLS YOU

The most interesting finding in this project is the one that partly breaks: H2's 2008 correlation number doesn't fit the panic story on its own. We didn't hide that or quietly drop 2008 – we said so, explained the likely reason (one company's unusual year

standing in for an entire region), and brought in H3 as an independent check. "The effect is real, but this one measurement of it is noisy" is a better, more honest finding than pretending every number lined up perfectly.

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What you can (and can't) prove with this file

You can credibly claim

- Co-movement and correlation between companies or regions
- Relative performance – who outran whom, and over what window
- Volatility and drawdowns – how rough the ride was
- The timing of when behavior changed
- Dispersion within a group – how much the members differ

You cannot claim

- That anything *caused* anything – there's no earnings or news data here
- Anything about revenue, valuation, or company quality
- Anything about management decisions or strategy
- Anything about what happens next

Timing and correlation breaks are your strongest evidence. When a chart shows something changed on a specific date, let the reader connect that to what they already know happened – don't assert the cause yourself.

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Where we are now

Assignment 3's draft is written, and it's built on exactly the H1/H2/H3 story above – nothing on this page is hypothetical anymore. Figure A (H1) and Figure B (H2/H3) are both real, executed charts in `assignment3_analysis.ipynb`, saved into `assignments/`, and embedded in the report itself.

Three things are still sitting as placeholders in the docx, waiting on the team rather than invented by anyone: our three names, the dataset's source link, and a proper read-through together before it's submitted.

After it's in: redo the H2/H3 check with two or three stocks per region instead of one – that's what should clean up the noisy 2008 correlation number flagged in section 11 – then start turning H1-H3 into the final chart set for the semester project.